

Customer Shopping Behaviour Analysis Report

1. Project Overview

This project examines customer shopping behaviour using transactional data from 3,900 purchases across multiple product categories. It aims to reveal patterns in spending, customer segments, product preferences, and subscription behavior, providing a data-driven basis for strategic business decisions.

2. Dataset summary

The dataset consists of 3900 rows and 18 columns. There are 37 missing values in the Review Rating column. Some key features include:

- Demographics - Customer ID, Age, Gender, Location
- Product details - Item Purchased, Category, Size, Color, Season
- Transaction Details - Purchase Amount (USD), Shipping Type, Payment Method
- Behavioural factors - Review Rating, Subscription Status, Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases

3. Exploratory Data Analysis using Python

I start with data preparation and cleaning in Python.

1. Data Loading: Import dataset using pandas.

2. Initial Exploration: Used [df.info\(\)](#) to check for structure and `df.describe()` for summary statistics.

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3900 entries, 0 to 3899
Data columns (total 18 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Customer ID                          3900 non-null   int64
1   Age                                   3900 non-null   int64
2   Gender                               3900 non-null   object
3   Item Purchased                       3900 non-null   object
4   Category                             3900 non-null   object
5   Purchase Amount (USD)                3900 non-null   int64
6   Location                             3900 non-null   object
7   Size                                 3900 non-null   object
8   Color                                3900 non-null   object
9   Season                               3900 non-null   object
10  Review Rating                        3863 non-null   float64
11  Subscription Status                  3900 non-null   object
12  Shipping Type                       3900 non-null   object
13  Discount Applied                    3900 non-null   object
14  Promo Code Used                     3900 non-null   object
15  Previous Purchases                   3900 non-null   int64
16  Payment Method                      3900 non-null   object
17  Frequency of Purchases               3900 non-null   object
dtypes: float64(1), int64(4), object(13)
```

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscripti Stat
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	39
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	28
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750065	N
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716983	N
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	N
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	N
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	N
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	N
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	N

3. Handle Missing Data: Checked for null values and inputted the missing values is the Review Rating column with the median rating of each product category.

4. Standardisation of Columns: Renamed columns to lower case, snake case names for later use in SQL and Power BI.

5. Feature Engineering:

- Created age_group column by binning customer ages.
- Created purchase_frequency_days column from purchase data.

6. Data Consistency Check: Verified if discount_applied and promo_code_used were redundant, promo_code_used dropped.

7. Data Integration: Connected Python script to PostgreSQL and loaded the cleaned dataframe for SQL analysis.

4. Data Analysis using SQL

I performed structured analysis in PostgreSQL to answer key business questions.

1. Revenue by gender - Compared total revenue generated by male and female customers.

	gender 	revenue 
1	Female	75191
2	Male	157890

2. High-spending discount users - Identified the customers who used a discount but still spent more than the average purchase amount.

	customer_id bigint	purchase_amount bigint
1	2	64
2	3	73
3	4	90
4	7	85
5	9	97
6	12	68
7	13	72
8	16	81
9	20	90
10	22	62
11	24	88
Total rows: 839		Query complete 00:00

3. Top 5 products by review ratings - Identified products with the highest average review ratings.

	item_purchased text	average product rating numeric
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.78

4. Shipping Type comparison - Compared the average purchase amounts between standard and express shipping.

	shipping_type text	round numeric
1	Standard	58.46
2	Express	60.48

5. Subscription type comparison - Compared the average spend and total revenue between subscribers and non-subscribers.

	subscription_status text	total_customers bigint	average_spend numeric	total_revenue numeric
1	No	2847	59.87	170436
2	Yes	1053	59.49	62645

6. Discount-dependent products - Identified top 5 products with the highest percentage of discounted purchases.

	item_purchased text	discount_rate numeric
1	Hat	50.00
2	Sneakers	49.66
3	Coat	49.07
4	Sweater	48.17
5	Pants	47.37

7. Customer segmentation - Classified customers into New, Returning, and Loyal segments based on their purchase history.

	customer_segment text	number of customers bigint
1	Loyal	3116
2	New	83
3	Returning	701

8. Top 3 most purchased products per category - Listed the most purchased products within each category.

	item_rank bigint 🔒	category text 🔒	item_purchased text 🔒	total_orders bigint 🔒
1	1	Accessori...	Jewelry	171
2	2	Accessori...	Sunglasses	161
3	3	Accessori...	Belt	161
4	1	Clothing	Blouse	171
5	2	Clothing	Pants	171
6	3	Clothing	Shirt	169
7	1	Footwear	Sandals	160
8	2	Footwear	Shoes	150
9	3	Footwear	Sneakers	145
10	1	Outerwear	Jacket	163
11	2	Outerwear	Coat	161

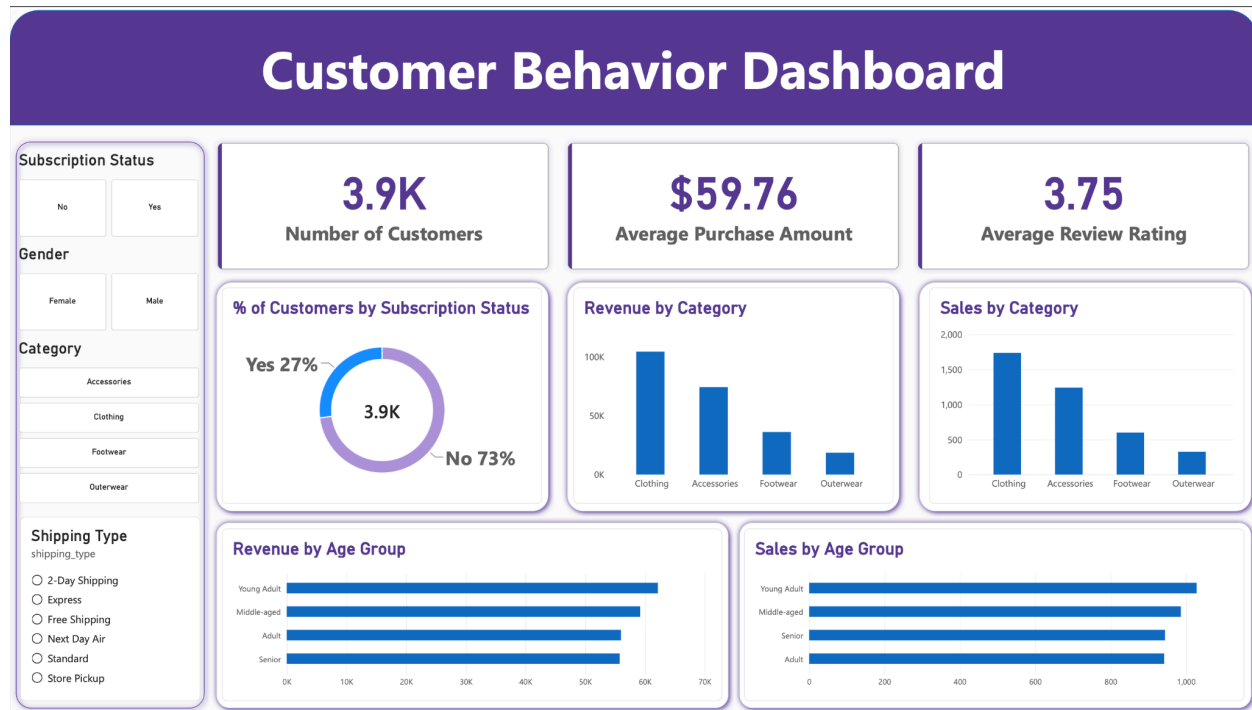
9. Repeat buyers and subscriptions - Checked whether customers with more than 5 purchases are more likely to subscribe.

	subscription_status text 🔒	repeat_buyers bigint 🔒
1	No	2518
2	Yes	958

10. Revenue by age group - Calculated the revenue contribution of each age group.

	age_group text 🔒	total_revenue numeric 🔒
1	Young Adult	62143
2	Middle-aged	59197
3	Adult	55978
4	Senior	55763

5. Dashboard in Power BI



6. Business Recommendations

- **Customer Loyalty Program** - Run a conversion campaign aimed at the 73%, such as a trial period, a first-order incentive, or perks that go beyond price (early access, free shipping thresholds). Track whether subscribers buy more often or spend more, so the program's value can be proven.
- **Focus on clothing and accessories** - Clothing and Accessories together generate roughly three quarters of revenue. Hence, we could invest in these two categories through stock depth, featured placements, and bundles that pair accessories with clothing purchases (for example, a belt or scarf suggested at checkout).
- **Improve on average review rating** - Fix issues in lowly rated best sellers and push sales of highly rated items through marketing.